# Linear Algebra Proofs

## Theorem 4.1.2: Properties of the Euclidean Inner Product

### Statement

> If $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ are vectors in $\mathbb{R}^n$ and $k$ is any scalar ($k \in \mathbb{R}$), then:
>
> - **(a) Symmetry Property:** $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
> - **(b) Additivity / Distributive Property:** $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$
> - **(c) Homogeneity / Scalar Property:** $(k\mathbf{u}) \cdot \mathbf{v} = k(\mathbf{u} \cdot \mathbf{v})$
> - **(d) Positivity Property:** $\mathbf{v} \cdot \mathbf{v} \geq 0$. Further, $\mathbf{v} \cdot \mathbf{v} = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

Let the vectors in $\mathbb{R}^n$ be denoted by their components:

$$\mathbf{u} = (u_1, u_2, \ldots, u_n), \quad \mathbf{v} = (v_1, v_2, \ldots, v_n), \quad \mathbf{w} = (w_1, w_2, \ldots, w_n)$$

By definition, the Euclidean inner product (dot product) is:

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + \cdots + u_nv_n$$

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### Proof of Part (a): $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ (Symmetry)

By definition of the Euclidean inner product:

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + \cdots + u_nv_n$$

Since multiplication of real numbers is commutative ($u_iv_i = v_iu_i$ for all real numbers $u_i, v_i$):

$$= v_1u_1 + v_2u_2 + \cdots + v_nu_n$$

$$= \mathbf{v} \cdot \mathbf{u}$$

$$\blacksquare$$

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### Proof of Part (b): $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$ (Additivity)

First, compute the vector addition:

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1,\ u_2 + v_2,\ \ldots,\ u_n + v_n)$$

Now, compute the inner product with $\mathbf{w}$:

$$(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = (u_1 + v_1)w_1 + (u_2 + v_2)w_2 + \cdots + (u_n + v_n)w_n$$

By the distributive property of real numbers ($(u_i + v_i)w_i = u_iw_i + v_iw_i$):

$$= (u_1w_1 + v_1w_1) + (u_2w_2 + v_2w_2) + \cdots + (u_nw_n + v_nw_n)$$

Regrouping terms by vector components:

$$= (u_1w_1 + u_2w_2 + \cdots + u_nw_n) + (v_1w_1 + v_2w_2 + \cdots + v_nw_n)$$

$$= \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$$

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## Proof of Part (c): $(k\mathbf{u}) \cdot \mathbf{v} = k(\mathbf{u} \cdot \mathbf{v})$ (Homogeneity)

First, compute scalar multiplication:

$$k\mathbf{u} = (ku_1,\ ku_2,\ \ldots,\ ku_n)$$

Now, compute the inner product with $\mathbf{v}$:

$$(k\mathbf{u}) \cdot \mathbf{v} = (ku_1)v_1 + (ku_2)v_2 + \cdots + (ku_n)v_n$$

By the associative and commutative properties of real multiplication ($(ku_i)v_i = k(u_iv_i)$):

$$= k(u_1v_1) + k(u_2v_2) + \cdots + k(u_nv_n)$$

Factoring out the common scalar $k$:

$$= k(u_1v_1 + u_2v_2 + \cdots + u_nv_n)$$

$$= k(\mathbf{u} \cdot \mathbf{v})$$

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## Proof of Part (d): $\mathbf{v} \cdot \mathbf{v} \geq 0$, and $\mathbf{v} \cdot \mathbf{v} = 0 \iff \mathbf{v} = \mathbf{0}$ (Positivity)

By definition of the Euclidean inner product:

$$\mathbf{v} \cdot \mathbf{v} = v_1v_1 + v_2v_2 + \cdots + v_nv_n = v_1^2 + v_2^2 + \cdots + v_n^2$$

1. **Non-negativity ($\mathbf{v} \cdot \mathbf{v} \geq 0$):** The square of any real number is non-negative ($v_i^2 \geq 0$ for all $i = 1, 2, \ldots, n$). Because the sum of non-negative real numbers is non-negative:

   $$\mathbf{v} \cdot \mathbf{v} = v_1^2 + v_2^2 + \cdots + v_n^2 \geq 0$$

2. **Zero condition ($\mathbf{v} \cdot \mathbf{v} = 0 \iff \mathbf{v} = \mathbf{0}$):** A sum of squares of real numbers equals zero:

   $$v_1^2 + v_2^2 + \cdots + v_n^2 = 0$$

   if and only if **every individual term is zero**:

   $$v_1 = 0, \quad v_2 = 0, \quad \ldots, \quad v_n = 0$$

   which holds if and only if:

   $$\mathbf{v} = (0, 0, \ldots, 0) = \mathbf{0}$$

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# Theorem 4.1.4: Properties of Length (Norm) in $\mathbb{R}^n$

## Statement

> If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$ and $k$ is any scalar ($k \in \mathbb{R}$), then:
>
> - **(a) Non-negativity:** $\|\mathbf{u}\| \geq 0$
> - **(b) Zero Condition:** $\|\mathbf{u}\| = 0$ if and only if $\mathbf{u} = \mathbf{0}$
> - **(c) Homogeneity:** $\|k\mathbf{u}\| = |k|\|\mathbf{u}\|$
> - **(d) Triangle Inequality:** $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

Let $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ and $\mathbf{v} = (v_1, v_2, \ldots, v_n)$. By definition, the Euclidean norm (length) of a vector is:

$$\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}} = \sqrt{u_1^2 + u_2^2 + \cdots + u_n^2}$$

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## Proof of Part (a): $\|\mathbf{u}\| \geq 0$

By definition:

$$\|\mathbf{u}\| = \sqrt{u_1^2 + u_2^2 + \cdots + u_n^2}$$

Since $u_i^2 \geq 0$ for each component $i = 1, \ldots, n$, the sum of squares is non-negative:

$$u_1^2 + u_2^2 + \cdots + u_n^2 \geq 0$$

The principal (non-negative) square root of a non-negative real number is always non-negative:

$$\|\mathbf{u}\| \geq 0$$

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## Proof of Part (b): $\|\mathbf{u}\| = 0 \iff \mathbf{u} = \mathbf{0}$

$$\|\mathbf{u}\| = 0 \iff \sqrt{u_1^2 + u_2^2 + \cdots + u_n^2} = 0$$

Squaring both sides:

$$\iff u_1^2 + u_2^2 + \cdots + u_n^2 = 0$$

Because each $u_i^2 \geq 0$, a sum of non-negative real numbers can only equal zero if every individual term is zero:

$$\iff u_1 = 0, \quad u_2 = 0, \quad \ldots, \quad u_n = 0$$

$$\iff \mathbf{u} = (0, 0, \ldots, 0) = \mathbf{0}$$

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## Proof of Part (c): $\|k\mathbf{u}\| = |k|\|\mathbf{u}\|$

Let $\mathbf{u} = (u_1, u_2, \ldots, u_n)$. The scaled vector is:

$$k\mathbf{u} = (ku_1, ku_2, \ldots, ku_n)$$

Applying the definition of the norm:

$$\|k\mathbf{u}\| = \sqrt{(ku_1)^2 + (ku_2)^2 + \cdots + (ku_n)^2}$$

$$= \sqrt{k^2u_1^2 + k^2u_2^2 + \cdots + k^2u_n^2}$$

Factor out the common term $k^2$:

$$= \sqrt{k^2(u_1^2 + u_2^2 + \cdots + u_n^2)}$$

Using the square root property $\sqrt{a \cdot b} = \sqrt{a} \cdot \sqrt{b}$, and noting that $\sqrt{k^2} = |k|$:

$$= \sqrt{k^2} \cdot \sqrt{u_1^2 + u_2^2 + \cdots + u_n^2}$$

$$= |k|\|\mathbf{u}\|$$

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## Proof of Part (d): $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ (The Triangle Inequality)

Start with the left-hand side squared using the inner product relationship $\|\mathbf{x}\|^2 = \mathbf{x} \cdot \mathbf{x}$:

$$\|\mathbf{u} + \mathbf{v}\|^2 = (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v})$$

Expanding using the distributive and symmetry properties of the dot product (Theorem 4.1.2):

$$= \mathbf{u} \cdot \mathbf{u} + \mathbf{u} \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{u} + \mathbf{v} \cdot \mathbf{v}$$

$$= \|\mathbf{u}\|^2 + 2(\mathbf{u} \cdot \mathbf{v}) + \|\mathbf{v}\|^2$$

By the properties of real numbers, any number is less than or equal to its absolute value ($x \leq |x|$):

$$\leq \|\mathbf{u}\|^2 + 2|\mathbf{u} \cdot \mathbf{v}| + \|\mathbf{v}\|^2$$

Now, apply the **Cauchy-Schwarz Inequality** ($|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\|\|\mathbf{v}\|$):

$$\leq \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}\| + \|\mathbf{v}\|^2$$

Notice the perfect square identity on the right side ($a^2 + 2ab + b^2 = (a + b)^2$):

$$= (\|\mathbf{u}\| + \|\mathbf{v}\|)^2$$

Putting the entire inequality chain together:

$$\|\mathbf{u} + \mathbf{v}\|^2 \leq (\|\mathbf{u}\| + \|\mathbf{v}\|)^2$$

Since both $\|\mathbf{u} + \mathbf{v}\|$ and $(\|\mathbf{u}\| + \|\mathbf{v}\|)$ are non-negative, taking the positive square root of both sides preserves the inequality:

$$\|\mathbf{u}+\mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$$

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# Theorem 4.1.5: Properties of Distance in $\mathbb{R}^n$

## Statement

> If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in $\mathbb{R}^n$ and $k$ is any scalar, then:
>
> - **(a) Non-negativity:** $d(\mathbf{u}, \mathbf{v}) \geq 0$
> - **(b) Zero Condition / Identity:** $d(\mathbf{u}, \mathbf{v}) = 0$ if and only if $\mathbf{u} = \mathbf{v}$
> - **(c) Symmetry:** $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$
> - **(d) Triangle Inequality:** $d(\mathbf{u}, \mathbf{v}) \leq d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$

Let $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ and $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ be vectors in $\mathbb{R}^n$. By definition, the Euclidean distance between $\mathbf{u}$ and $\mathbf{v}$ is given by the norm of their difference:

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \cdots + (u_n - v_n)^2}$$

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## Proof of Part (a): $d(\mathbf{u}, \mathbf{v}) \geq 0$ (Non-negativity)

**Method 1 (Using Norm Properties):** By definition of distance:

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

Let $\mathbf{x} = \mathbf{u} - \mathbf{v}$. By Theorem 4.1.4(a), the norm of any vector in $\mathbb{R}^n$ is non-negative:

$$\|\mathbf{x}\| \geq 0 \implies \|\mathbf{u} - \mathbf{v}\| \geq 0$$

Therefore:

$$d(\mathbf{u}, \mathbf{v}) \geq 0$$

**Method 2 (Component-wise):**

$$d(\mathbf{u}, \mathbf{v}) = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \cdots + (u_n - v_n)^2}$$

Since $(u_i - v_i)^2 \geq 0$ for each $i = 1, 2, \ldots, n$, the sum of squares is non-negative:

$$\sum_{i=1}^{n} (u_i - v_i)^2 \geq 0$$

Because the principal square root of a non-negative number is non-negative:

$$d(\mathbf{u}, \mathbf{v}) \geq 0$$

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## Proof of Part (b): $d(\mathbf{u}, \mathbf{v}) = 0 \iff \mathbf{u} = \mathbf{v}$ (Zero Condition)

**Method 1 (Using Norm Properties):** By definition:

$$d(\mathbf{u}, \mathbf{v}) = 0 \iff \|\mathbf{u} - \mathbf{v}\| = 0$$

By Theorem 4.1.4(b), the norm of a vector is zero if and only if the vector itself is the zero vector:

$$\|\mathbf{u} - \mathbf{v}\| = 0 \iff \mathbf{u} - \mathbf{v} = \mathbf{0}$$

Adding $\mathbf{v}$ to both sides:

$$\iff \mathbf{u} = \mathbf{v}$$

**Method 2 (Component-wise):**

$$d(\mathbf{u}, \mathbf{v}) = 0 \iff \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \cdots + (u_n - v_n)^2} = 0$$

Squaring both sides:

$$\iff (u_1 - v_1)^2 + (u_2 - v_2)^2 + \cdots + (u_n - v_n)^2 = 0$$

A sum of non-negative squares equals zero if and only if every term is zero:

$$\iff (u_i - v_i)^2 = 0 \quad \text{for all } i = 1, 2, \ldots, n$$

$$\iff u_i - v_i = 0 \iff u_i = v_i \quad \text{for all } i = 1, 2, \ldots, n$$

$$\iff \mathbf{u} = \mathbf{v}$$

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## Proof of Part (c): $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$ (Symmetry)

**Method 1 (Using Norm Homogeneity):** By definition:

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

Factor out $-1$ from the vector difference:

$$\mathbf{u} - \mathbf{v} = -(\mathbf{v} - \mathbf{u}) = (-1)(\mathbf{v} - \mathbf{u})$$

Using Theorem 4.1.4(c) ($\|k\mathbf{x}\| = |k|\|\mathbf{x}\|$ with $k = -1$):

$$\|(-1)(\mathbf{v} - \mathbf{u})\| = |-1|\|\mathbf{v} - \mathbf{u}\| = 1 \cdot \|\mathbf{v} - \mathbf{u}\| = \|\mathbf{v} - \mathbf{u}\|$$

Since $\|\mathbf{v} - \mathbf{u}\| = d(\mathbf{v}, \mathbf{u})$:

$$d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$$

**Method 2 (Component-wise):** In each component of the sum:

$$(u_i - v_i)^2 = [-(v_i - u_i)]^2 = (-1)^2(v_i - u_i)^2 = (v_i - u_i)^2$$

Thus:

$$d(\mathbf{u},\mathbf{v}) = \sqrt{\sum_{i=1}^{n}(u_i - v_i)^2} = \sqrt{\sum_{i=1}^{n}(v_i - u_i)^2} = d(\mathbf{v},\mathbf{u})$$

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### Proof of Part (d): $d(\mathbf{u},\mathbf{v}) \leq d(\mathbf{u},\mathbf{w}) + d(\mathbf{w},\mathbf{v})$ (The Triangle Inequality for Distance)

By definition:

$$d(\mathbf{u},\mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

Add and subtract the third vector $\mathbf{w}$ inside the difference:

$$\mathbf{u} - \mathbf{v} = (\mathbf{u} - \mathbf{w}) + (\mathbf{w} - \mathbf{v})$$

Taking the norm of both sides:

$$d(\mathbf{u},\mathbf{v}) = \|(\mathbf{u} - \mathbf{w}) + (\mathbf{w} - \mathbf{v})\|$$

Let $\mathbf{a} = \mathbf{u} - \mathbf{w}$ and $\mathbf{b} = \mathbf{w} - \mathbf{v}$. By the Triangle Inequality for vector norms (Theorem 4.1.4(d), $\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\|$):

$$\|(\mathbf{u} - \mathbf{w}) + (\mathbf{w} - \mathbf{v})\| \leq \|\mathbf{u} - \mathbf{w}\| + \|\mathbf{w} - \mathbf{v}\|$$

By definition of distance:

$$\|\mathbf{u} - \mathbf{w}\| = d(\mathbf{u},\mathbf{w}) \quad \text{and} \quad \|\mathbf{w} - \mathbf{v}\| = d(\mathbf{w},\mathbf{v})$$

Substituting these back into the inequality:

$$d(\mathbf{u},\mathbf{v}) \leq d(\mathbf{u},\mathbf{w}) + d(\mathbf{w},\mathbf{v})$$

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### Key Geometric Takeaways:

- $d(\mathbf{u},\mathbf{v})$ represents the straight-line Euclidean distance between two points.
- Any detour via a third point $\mathbf{w}$ can never be shorter than the straight-line segment: $d(\mathbf{u},\mathbf{v}) \leq d(\mathbf{u},\mathbf{w}) + d(\mathbf{w},\mathbf{v})$.
- Equality holds if and only if $\mathbf{w}$ lies directly on the straight-line segment between $\mathbf{u}$ and $\mathbf{v}$.

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## Theorem 4.1.6: The Polarization Identity in $\mathbb{R}^n$

### Statement

> If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$ with the Euclidean inner product, then:
>
> $$\mathbf{u} \cdot \mathbf{v} = \frac{1}{4}\|\mathbf{u} + \mathbf{v}\|^2 - \frac{1}{4}\|\mathbf{u} - \mathbf{v}\|^2$$

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What is this and why does it matter?

- **Reconstructing the Dot Product from Lengths alone:** Notice that the left-hand side is the **inner product $\mathbf{u} \cdot \mathbf{v}$** (which determines angles and orthogonality). The right-hand side contains **only norms (lengths)** of vectors ($\|\mathbf{u}+\mathbf{v}\|$ and $\|\mathbf{u}-\mathbf{v}\|$). This proves that **if you only have a tool to measure lengths, you can still compute angles and dot products!**
- **High School Algebra Connection:** This is the vector analog of the familiar binomial product formula:

  $$xy = \frac{1}{4}(x+y)^2 - \frac{1}{4}(x-y)^2$$

  where squaring $(x+y)$ and $(x-y)$ produces cross-terms $2xy$ and $-2xy$, whose difference leaves $4xy$.

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Proof

First, expand the squared norm of the sum $\|\mathbf{u}+\mathbf{v}\|^2$ using the relationship between norm and inner product ($\|\mathbf{x}\|^2 = \mathbf{x} \cdot \mathbf{x}$):

$$\|\mathbf{u}+\mathbf{v}\|^2 = (\mathbf{u}+\mathbf{v}) \cdot (\mathbf{u}+\mathbf{v})$$

Applying the distributive and symmetry properties of the Euclidean inner product (Theorem 4.1.2):

$$\|\mathbf{u}+\mathbf{v}\|^2 = \mathbf{u} \cdot \mathbf{u} + 2(\mathbf{u} \cdot \mathbf{v}) + \mathbf{v} \cdot \mathbf{v}$$

Using $\|\mathbf{u}\|^2 = \mathbf{u} \cdot \mathbf{u}$ and $\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}$:

$$\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2(\mathbf{u} \cdot \mathbf{v}) + \|\mathbf{v}\|^2 \quad \text{— (Equation 1)}$$

Next, expand the squared norm of the difference $\|\mathbf{u}-\mathbf{v}\|^2$:

$$\|\mathbf{u}-\mathbf{v}\|^2 = (\mathbf{u}-\mathbf{v}) \cdot (\mathbf{u}-\mathbf{v})$$

$$= \mathbf{u} \cdot \mathbf{u} - 2(\mathbf{u} \cdot \mathbf{v}) + \mathbf{v} \cdot \mathbf{v}$$

$$\|\mathbf{u}-\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2(\mathbf{u} \cdot \mathbf{v}) + \|\mathbf{v}\|^2 \quad \text{— (Equation 2)}$$

Now, subtract **Equation 2** from **Equation 1**:

$$\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 = \left(\|\mathbf{u}\|^2 + 2(\mathbf{u} \cdot \mathbf{v}) + \|\mathbf{v}\|^2\right) - \left(\|\mathbf{u}\|^2 - 2(\mathbf{u} \cdot \mathbf{v}) + \|\mathbf{v}\|^2\right)$$

Notice that the squared terms $\|\mathbf{u}\|^2$ and $\|\mathbf{v}\|^2$ cancel out:

$$\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 = 2(\mathbf{u} \cdot \mathbf{v}) - (-2(\mathbf{u} \cdot \mathbf{v}))$$

$$\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 = 4(\mathbf{u} \cdot \mathbf{v})$$

Divide both sides by $4$:

$$\mathbf{u} \cdot \mathbf{v} = \frac{1}{4}\|\mathbf{u}+\mathbf{v}\|^2 - \frac{1}{4}\|\mathbf{u}-\mathbf{v}\|^2$$

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### Related Identity: The Parallelogram Equation

What happens if you **add** Equations (1) and (2) instead of subtracting them?

$$\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$$

- **Geometric meaning:** The sum of the squares of the lengths of the diagonals of a parallelogram equals the sum of the squares of the lengths of its four sides.

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## Definition: Orthogonality in $\mathbb{R}^n$

> Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are called **orthogonal** (denoted $\mathbf{u} \perp \mathbf{v}$) if their Euclidean inner product is zero:
>
> $$\mathbf{u} \cdot \mathbf{v} = 0$$

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## Theorem 4.1.7: Pythagorean Theorem in $\mathbb{R}^n$

### Statement

> If $\mathbf{u}$ and $\mathbf{v}$ are orthogonal vectors in $\mathbb{R}^n$ with the Euclidean inner product ($\mathbf{u} \cdot \mathbf{v} = 0$), then:
>
> $$\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

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### Proof

Start by expressing the left-hand side using the relationship between vector norm and inner product ($\|\mathbf{x}\|^2 = \mathbf{x} \cdot \mathbf{x}$):

$$\|\mathbf{u}+\mathbf{v}\|^2 = (\mathbf{u}+\mathbf{v}) \cdot (\mathbf{u}+\mathbf{v})$$

Expand the dot product using the distributive property (Theorem 4.1.2(b)):

$$= \mathbf{u} \cdot (\mathbf{u}+\mathbf{v}) + \mathbf{v} \cdot (\mathbf{u}+\mathbf{v})$$

$$= \mathbf{u} \cdot \mathbf{u} + \mathbf{u} \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{u} + \mathbf{v} \cdot \mathbf{v}$$

Using the symmetry property ($\mathbf{v} \cdot \mathbf{u} = \mathbf{u} \cdot \mathbf{v}$) and the definition of norm ($\mathbf{u} \cdot \mathbf{u} = \|\mathbf{u}\|^2$ and $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$):

$$= \|\mathbf{u}\|^2 + 2(\mathbf{u} \cdot \mathbf{v}) + \|\mathbf{v}\|^2$$

Since $\mathbf{u}$ and $\mathbf{v}$ are given to be **orthogonal**, by definition $\mathbf{u} \cdot \mathbf{v} = 0$:

$$= \|\mathbf{u}\|^2 + 2(0) + \|\mathbf{v}\|^2$$

$$= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

Therefore:

$$\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

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Key Takeaways & Exam Tips:

1. **Geometric Connection to Right Triangles:** In $\mathbb{R}^2$ or $\mathbb{R}^3$, $\mathbf{u}$ and $\mathbf{v}$ form the perpendicular legs of a right triangle, while $\mathbf{u} + \mathbf{v}$ (or $\mathbf{u} - \mathbf{v}$) forms the hypotenuse:
$$(\text{hypotenuse})^2 = (\text{leg}_1)^2 + (\text{leg}_2)^2$$
2. **Difference of Vectors ($\|\mathbf{u} - \mathbf{v}\|^2$):** Similarly, for orthogonal vectors:
$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2(\mathbf{u} \cdot \mathbf{v}) + \|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$
Hence, if $\mathbf{u} \perp \mathbf{v}$, the diagonals of the rectangle spanned by $\mathbf{u}$ and $\mathbf{v}$ have equal length:
$$\|\mathbf{u} + \mathbf{v}\| = \|\mathbf{u} - \mathbf{v}\|$$
3. **Generalization (Extension to $k$ mutually orthogonal vectors):** If $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is a set of mutually orthogonal vectors ($\mathbf{v}_i \cdot \mathbf{v}_j = 0$ for $i \neq j$), then:
$$\|\mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_k\|^2 = \|\mathbf{v}_1\|^2 + \|\mathbf{v}_2\|^2 + \cdots + \|\mathbf{v}_k\|^2$$

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# Theorem 5.2.2: Solution Space of a Homogeneous System is a Subspace

## Statement

> If $A\mathbf{x} = \mathbf{0}$ is a homogeneous linear system of $m$ equations in $n$ unknowns, then the set of solution vectors is a subspace of $\mathbb{R}^n$.

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## Proof

Let $W$ be the set of all solution vectors to the homogeneous system:

$$W = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}\}$$

To prove that $W$ is a subspace of $\mathbb{R}^n$, we test the subspace conditions:

**1. Non-empty / Contains the Zero Vector ($\mathbf{0} \in W$)**

There is at least one vector in $W$, namely the zero vector $\mathbf{0} \in \mathbb{R}^n$, because:

$$A\mathbf{0} = \mathbf{0}$$

Since the zero vector satisfies the homogeneous equation, $\mathbf{0} \in W$, meaning $W$ is non-empty.

**2. Closure under Vector Addition**

Let $\mathbf{x}$ and $\mathbf{x}'$ be any two solution vectors in $W$. By definition:

$$A\mathbf{x} = \mathbf{0} \quad \text{and} \quad A\mathbf{x}' = \mathbf{0}$$

Now, test their sum $(\mathbf{x} + \mathbf{x}')$ by multiplying by $A$, using the distributive property of matrix multiplication:

$$A(\mathbf{x} + \mathbf{x}') = A\mathbf{x} + A\mathbf{x}' = \mathbf{0} + \mathbf{0} = \mathbf{0}$$

Since $A(\mathbf{x} + \mathbf{x}') = \mathbf{0}$, the vector $(\mathbf{x} + \mathbf{x}')$ is also a solution, meaning:

$$(\mathbf{x} + \mathbf{x}') \in W$$

Thus, $W$ is **closed under addition**.

**3. Closure under Scalar Multiplication**

Let $\mathbf{x} \in W$ (so $A\mathbf{x} = \mathbf{0}$) and let $k$ be any real scalar ($k \in \mathbb{R}$). Test the scaled vector $(k\mathbf{x})$ by multiplying by $A$:

$$A(k\mathbf{x}) = k(A\mathbf{x}) = k(\mathbf{0}) = \mathbf{0}$$

Since $A(k\mathbf{x}) = \mathbf{0}$, the vector $k\mathbf{x}$ is also a solution, meaning:

$$k\mathbf{x} \in W$$

Thus, $W$ is **closed under scalar multiplication**.

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### Conclusion

Since $W$ is non-empty and closed under both vector addition and scalar multiplication, by Theorem 5.2.1 (The Subspace Test), $W$ **is a valid subspace of** $\mathbb{R}^n$.

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### Key Takeaways for Exams:

1. **Where does $W$ live?** Because matrix $A$ is $m \times n$, the vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ has $n$ entries. Therefore, $W$ is a subspace of $\mathbb{R}^n$ (not $\mathbb{R}^m$).
2. **What if the system is non-homogeneous ($A\mathbf{x} = \mathbf{b}$ where $\mathbf{b} \neq \mathbf{0}$)?**
   - $\mathbf{0}$ is not a solution ($A\mathbf{0} = \mathbf{0} \neq \mathbf{b}$).
   - Closure fails: $A(\mathbf{x} + \mathbf{x}') = \mathbf{b} + \mathbf{b} = 2\mathbf{b} \neq \mathbf{b}$.
   - Therefore, the solution set to $A\mathbf{x} = \mathbf{b}$ is **NEVER a subspace** when $\mathbf{b} \neq \mathbf{0}$.

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## Theorem: Consistency Theorem for Linear Systems ($A\mathbf{x} = \mathbf{b}$)

### Statement

> A system of linear equations $A\mathbf{x} = \mathbf{b}$ is **consistent** if and only if the vector $\mathbf{b}$ is in the column space of $A$:

$$A\mathbf{x} = \mathbf{b} \text{ is consistent } \iff \mathbf{b} \in \text{Col}(A)$$

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## Core Intuition: The Column View of Matrix Multiplication

When you multiply an $m \times n$ matrix $A$ by a column vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$, the resulting vector is **not just a collection of dot products**—it is fundamentally a **linear combination of the columns of** $A$:

$$A\mathbf{x} = \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{c}_1 + x_2\mathbf{c}_2 + \cdots + x_n\mathbf{c}_n$$

The entries of $\mathbf{x}$ act as the **weights** scaling each column of $A$.

Therefore, asking *"Does $A\mathbf{x} = \mathbf{b}$ have a solution?"* is mathematically identical to asking *"Can vector $\mathbf{b}$ be built by mixing the columns of $A$?"*

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## Proof (Bidirectional $\iff$)

Let $A$ be an $m \times n$ matrix partitioned into its column vectors:

$$A = \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_n \end{bmatrix}, \quad \text{where each } \mathbf{c}_j \in \mathbb{R}^m$$

and let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$ and $\mathbf{b} \in \mathbb{R}^m$.

By the definition of matrix-vector multiplication:

$$A\mathbf{x} = x_1\mathbf{c}_1 + x_2\mathbf{c}_2 + \cdots + x_n\mathbf{c}_n \quad \text{— (Equation 1)}$$

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**1. Forward Direction ($\implies$): Assume $A\mathbf{x} = \mathbf{b}$ is consistent**

1. If the system $A\mathbf{x} = \mathbf{b}$ is consistent, there exists at least one solution vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ such that:

$$A\mathbf{x} = \mathbf{b}$$

2. Substituting Equation (1) into this equality:

$$x_1\mathbf{c}_1 + x_2\mathbf{c}_2 + \cdots + x_n\mathbf{c}_n = \mathbf{b}$$

3. This equation explicitly shows that $\mathbf{b}$ is expressed as a linear combination of the column vectors $\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_n$.
4. By definition, the **column space** $\mathrm{Col}(A)$ is the span of all column vectors:

$$\mathrm{Col}(A) = \mathrm{span}\{\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_n\}$$

5. Since $\mathbf{b}$ is a linear combination of these columns:

$$\mathbf{b} \in \mathrm{Col}(A)$$

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**2. Backward Direction ( $\Longleftarrow$ ): Assume $\mathbf{b} \in \mathrm{Col}(A)$**

1. If $\mathbf{b} \in \mathrm{Col}(A)$, then by definition of column space, $\mathbf{b}$ can be written as a linear combination of the columns of $A$.
2. That is, there exist scalars $s_1, s_2, \ldots, s_n \in \mathbb{R}$ such that:

$$s_1\mathbf{c}_1 + s_2\mathbf{c}_2 + \cdots + s_n\mathbf{c}_n = \mathbf{b}$$

3. Define the vector $\mathbf{s} \in \mathbb{R}^n$ by:

$$\mathbf{s} = \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{bmatrix}$$

4. Multiplying $A$ by $\mathbf{s}$ using Equation (1):

$$A\mathbf{s} = s_1\mathbf{c}_1 + s_2\mathbf{c}_2 + \cdots + s_n\mathbf{c}_n = \mathbf{b}$$

5. Since we found an explicit vector $\mathbf{s}$ that satisfies the equation $A\mathbf{s} = \mathbf{b}$, the linear system has a solution.
6. Therefore, the system $A\mathbf{x} = \mathbf{b}$ is **consistent**.

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Key Exam Takeaways:

1. **The "Range" of a Matrix:** $\mathrm{Col}(A)$ is the output space of the transformation $T(\mathbf{x}) = A\mathbf{x}$. You can only solve for $\mathbf{b}$ if $\mathbf{b}$ lands inside the output space.
2. **Rank Connection (Augmented Matrix):** $A\mathbf{x} = \mathbf{b}$ is consistent $\iff \mathrm{rank}(A) = \mathrm{rank}([A \mid \mathbf{b}])$. *(Adding $\mathbf{b}$ to the matrix doesn't add any new dimensions because $\mathbf{b}$ is already inside the span of $A$'s columns).*